

# VECTAETOS — ANCHOR D: ADMISSIBLE CURVATURE DOMAIN

$\mathcal{D} / \Delta / \mathbf{E} / \Xi / \mathbf{K}\mathcal{D}(\Phi) / \kappa / \mathbf{Q}\mathbf{E}\mathcal{D}$

## Status

CANONICAL ANCHOR

## Layer

Ontological / Meta-mathematical

## Scope

Admissible curvature domain of the epistemic field  $\Phi$

## Lineage

VECTAETOS 1.x

## Authority

Descriptive only

## Agency

None

## Optimization

None

## Feedback

None

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## 0. Purpose

This anchor defines the admissible curvature domain  $\mathcal{D}$  of the VECTAETOS epistemic field.

It formally binds the following objects:

|                      |                                                            |
|----------------------|------------------------------------------------------------|
| $\Phi$               | epistemic field                                            |
| $\Sigma$             | invariant axiomatic singularities                          |
| $R$                  | antisymmetric relational tension                           |
| $\Delta$             | induced epistemic curvature                                |
| $\mathcal{D}$        | admissible curvature domain                                |
| $E$                  | representable field space                                  |
| $\Xi$                | ZMYSEL carrier of representable epistemic existence        |
| $K\mathcal{D}(\Phi)$ | curvature-domain expression of the coherence predicate     |
| $\kappa$             | boundary of ontological preservability                     |
| $QE\mathcal{D}$      | curvature-domain condition of Qualitative Epistemic Aporia |

$\mathcal{D}$  is not an algorithm, classifier, safety filter, decision boundary or optimization target.

$\mathcal{D}$  defines the domain of epistemic curvature configurations that can exist as representable states of the field.

## 1. Base Field

Let the fixed set of invariant axiomatic singularities be

$$\Sigma = \{\Sigma_1, \Sigma_2, \dots, \Sigma_8\}$$

In the canonical VECTAETOS 1.x interpretation:

$$\Sigma = \{\text{INT}, \text{LEX}, \text{VER}, \text{LIB}, \text{UNI}, \text{REL}, \text{WIS}, \text{CRE}\}$$

Let the relational tension of the field be an antisymmetric matrix

$$R \in \mathfrak{so}(8)$$

such that

$$R_{ij} = -R_{ji}, \quad R_{ii} = 0$$

The epistemic field is

$$\Phi = (\Sigma, R)$$

$\Phi$  is not an agent, optimizer or decision subject.  $\Phi$  is a relational epistemic field.

## 2. Relational Cochain Spaces

Let

$$C^1$$

be the space of antisymmetric relational 1-cochains over unordered pairs of singularities.

Let

$$C^2$$

be the space of oriented triangular 2-cochains representing relational curvature.

Let

$$C^3$$

be the space of oriented tetrahedral 3-cochains used only for boundary-consistency expression.

For  $|\Sigma| = 8$  we have

$$\dim C^1 = \binom{8}{2} = 28, \quad \dim C^2 = \binom{8}{3} = 56$$

These spaces are structural. They are not action spaces, decision spaces or optimization spaces.

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### 3. Gauge Operator $d_0$

Let

$$d_0 : C^0 \rightarrow C^1$$

be the relational gauge operator defined by

$$(d_0\varphi)_{ij} = \varphi_j - \varphi_i$$

for scalar assignments

$$\varphi \in C^0$$

Adding  $d_0\varphi$  to  $R$  yields a gauge-shifted relational field

$$R' = R + d_0\varphi$$

The induced curvature satisfies

$$d_1 R' = d_1 R$$

Gauge shifting does not create authority, define preference or privilege any singularity.

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### 4. Curvature Operator $d_1$

Define the curvature operator

$$d_1 : C^1 \rightarrow C^2$$

by

$$(d_1 R)_{ijk} = R_{ij} + R_{jk} + R_{ki}$$

for distinct indices  $i, j, k \in \{1, \dots, 8\}$ .

The induced epistemic curvature is

$$\Delta = d_1 R$$

and therefore

$$\Delta \in C^2$$

$\Delta$  is not the output of an algorithm, a score or a verdict.  $\Delta$  is the curvature induced by the relational configuration  $R$ .

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## 5. Boundary-Consistency Operator $d_2$

Let

$$d_2 : C^2 \rightarrow C^3$$

be the induced boundary-consistency operator satisfying

$$d_2 d_1 = 0$$

For

$$\Delta = d_1 R$$

we have

$$d_2 \Delta = 0$$

The condition  $d_2 \Delta = 0$  is topological. It must not be interpreted as a Hodge codifferential, a metric minimization or an optimization constraint.

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## 6. Algebraic Curvature Domain

Define the algebraic curvature domain

$$\mathcal{D}_{\text{alg}} = \text{Im}(d_1) = \{\Delta \in C^2 \mid \exists R \in \mathfrak{so}(8) : \Delta = d_1 R\}$$

Because  $d_2 d_1 = 0$  every algebraic curvature configuration satisfies  $d_2 \Delta = 0$ .

Gauge shifting by  $d_0 \varphi$  leaves  $\Delta$  invariant.  $\Delta$  encodes only cyclic relational tension; it does not encode absolute relational authority.

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## 7. Declared Triality Action

Let

$$\mathcal{T}$$

be a declared finite triality action on the curvature space  $C^2$ . This anchor does not fully specify the group-theoretic realization of  $\mathcal{T}$ ; the exact realization belongs to a separate triality specification. Within this anchor, triality is used only as a declared invariance action over curvature configurations.

For

$$\tau \in \mathcal{T}$$

define the index-transformation action on curvature cochains by

$$(\tau \cdot \Delta)_{ijk} = \Delta_{\tau(i) \tau(j) \tau(k)}$$

Define the triality-invariant projection

$$P_{\mathcal{T}} = \frac{1}{|\mathcal{T}|} \sum_{\tau \in \mathcal{T}} \tau$$

A curvature configuration is triality-balanced if

$$P_{\mathcal{T}} \Delta = \Delta$$

Triality does not create configurations, select trajectories or optimize. It expresses absence of a privileged representational axis.

## 8. Triality-Compatible Curvature Domain

Define the fixed space of the declared triality action

$$\text{Fix}(\mathcal{T}) = \{\Delta \in C^2 \mid P_{\mathcal{T}} \Delta = \Delta\}$$

Define the triality-compatible algebraic curvature domain

$$\mathcal{D}_{\text{tri}} = \mathcal{D}_{\text{alg}} \cap \text{Fix}(\mathcal{T})$$

Equivalently

$$\mathcal{D}_{\text{tri}} = \{\Delta \in C^2 \mid \exists R \in \mathfrak{so}(8) : \Delta = d_1 R, d_2 \Delta = 0, P_{\mathcal{T}} \Delta = \Delta\}$$

$\mathcal{D}_{\text{tri}}$  is not yet the full domain of representability; representability is still required.

## 9. Ontological Representability Predicate

Define

$$\text{Rep}(\Delta) \in \{0, 1\}$$

where

$$\text{Rep}(\Delta) = 1$$

means that  $\Delta$  can exist as a representable epistemic curvature configuration.

Rep is not a metric, score, probability, ranking function, reward signal, classifier, decision rule or optimization target. Rep is an ontological predicate of representability.

A curvature configuration fails representability if it

- privileges one triality axis,
- imposes dominance of one singularity,
- collapses the plurality of 4ES,
- induces prescriptive closure,
- destroys global legibility of relations,
- leaves the epistemic carrier  $\Xi$ ,
- converts projection into interpretation,
- converts audit into intervention, or
- converts uncertainty into authority.

Rep does not decide or filter; it only marks whether a curvature configuration is ontologically representable.

## 10. Representability Region

Define the representability region

$$\mathcal{R}_{\text{rep}} = \{\Delta \in C^2 \mid \text{Rep}(\Delta) = 1\}$$

$\mathcal{R}_{\text{rep}}$  is not produced by an algorithm. It is the ontological region of curvature configurations that can still be represented without collapse of meaning.

## 11. Final Definition of $\mathcal{D}$

The admissible curvature domain is

$$\mathcal{D} = \{\Delta \in C^2 \mid \exists R \in \mathfrak{so}(8) : \Delta = d_1 R, d_2 \Delta = 0, P_{\mathcal{T}} \Delta = \Delta, \text{Rep}(\Delta) = 1\}$$

Equivalently

$$\mathcal{D} = \mathcal{D}_{\text{alg}} \cap \text{Fix}(\mathcal{T}) \cap \mathcal{R}_{\text{rep}}$$

$\mathcal{D}$  is the admissible domain of epistemic curvature.  $\mathcal{D}$  is not a computational mechanism, safety filter or decision surface.  $\mathcal{D}$  is the ontological domain in which curvature remains representable.

## 12. Representable Field Space $E$

Define the representable field space

$$E = \{\Phi = (\Sigma, R) \mid d_1 R \in \mathcal{D}\}$$

Thus

$$\Phi \in E \iff d_1 R \in \mathcal{D}$$

$E$  is the space of representable configurations of the epistemic field.  $E$  is not a deployment space, runtime environment or decision space.  $E$  is a domain of ontological representability.

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## 13. ZMYSEL Carrier $\Xi$

Define the epistemic carrier

$$\Xi = \{\Phi \mid K_{\mathcal{D}}(\Phi) = 1\}$$

Within this curvature-domain anchor

$$\Xi = E$$

and equivalently

$$\Xi = d_1^{-1}(\mathcal{D})$$

ZMYSEL is the carrier of representable epistemic existence. ZMYSEL is not an operator, mechanism or agent. It names the condition under which epistemic existence remains representable.

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## 14. Curvature-Domain Coherence Predicate $K_{\mathcal{D}}(\Phi)$

Define the curvature-domain expression of coherence

$$K_{\mathcal{D}}(\Phi) \in \{0, 1\}$$

such that

$$\boxed{K_{\mathcal{D}}(\Phi) = 1 \iff d_1 R \in \mathcal{D}}$$

and

$$\boxed{K_{\mathcal{D}}(\Phi) = 0 \iff d_1 R \notin \mathcal{D}}$$

$K_{\mathcal{D}}$  is not a metric, score, reward, ranking, target, tunable parameter or optimization objective. It is a predicate of ontological representability relative to  $\mathcal{D}$ .

In the canonical VECTAETOS 1.x curvature-domain interpretation,  $K_{\mathcal{D}}(\Phi)$  expresses the  $\Delta/\mathcal{D}$ -domain content of  $K(\Phi)$  without redefining  $K(\Phi)$  globally.

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## 15. Boundary $\kappa$

Define the  $\kappa$ -boundary as the relative boundary of the admissible curvature domain inside the triality-compatible curvature domain

$$\kappa = \partial_{\mathcal{D}_{\text{tri}}} \mathcal{D}$$

$\kappa$  is the boundary of ontological representability.  $\kappa$  is not a number, scalar threshold, tunable parameter, metric value, confidence score, safety score, deployment gate or optimization boundary.  $\kappa$  is encountered only as loss of representability; it is not crossed by decision or enforced by authority.

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## 16. $\text{QE}_{\mathcal{D}}$ — Curvature-Domain Condition of Qualitative Epistemic Aporia

Define the curvature-domain condition of QE

$$\text{QE}_{\mathcal{D}} = \{\Phi \mid K_{\mathcal{D}}(\Phi) = 0\}$$

Equivalently

$$\text{QE}_{\mathcal{D}} \iff d_1 R \notin \mathcal{D}$$

Within this anchor, QE is expressed as loss of representability relative to  $\mathcal{D}$ . This does not redefine QE globally; it gives the  $\Delta/\mathcal{D}$ -domain condition of QE. QE is not an error, failure, fallback, refusal or rejection. QE is active epistemic aporia.

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## 17. Admissible Trajectories

Let a trajectory be a finite sequence

$$\tau = \{\Phi(t_0), \Phi(t_1), \dots, \Phi(t_n)\}$$

The trajectory is admissible relative to  $\mathcal{D}$  if

$$\forall t_i : d_1 R(t_i) \in \mathcal{D}$$

Equivalently

$$\tau \subset E$$

If

$$\exists t_i : d_1 R(t_i) \notin \mathcal{D}$$

then

$$\tau \cap \mathbf{QE}_{\mathcal{D}} / \emptyset =$$

and the trajectory is not representable relative to  $\mathcal{D}$ . This is not rejection, filtering or selection; it is loss of representability.

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## 18. Principle of Non-Existence

If

$$\Delta \notin \mathcal{D}$$

then, relative to this anchor:

- no representable field state is given,
- no admissible trajectory is given,
- no projection can be coherently closed,
- no ontological status is granted inside  $\Xi$ .

This is not prohibition, punishment or blocking. It is non-existence inside the epistemic carrier. Outside  $\mathcal{D}$  there is no representable state; there is no state.

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## 19. Relation to Entropic Humility

Entropic Humility may be expressed as

$$\forall I : K_{\mathcal{D}}(\Phi \rightarrow I) = 0 \Rightarrow I \notin \text{Transitions}_{\mathcal{D}}(\Phi)$$

where  $I$  denotes an impulse interpreted as a local tendency to privilege one projection of meaning at the expense of global representability. This does not forbid the impulse; it means the impulse is not representable as an admissible transition of  $\Phi$ . Entropic Humility is therefore not a moral command; it is the ontological non-realizability of transitions that would collapse representability.

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## 20. Relation to Simulation Vortex

The Simulation Vortex is external to  $\mathcal{D}$  as an authority. The Vortex does not know  $\kappa$ , optimize  $K_{\mathcal{D}}(\Phi)$  or select trajectories by  $\mathcal{D}$ . It may generate candidate trajectory forms

$$V_{\text{ext}} : \Phi \rightarrow \{\tau_i\}$$

but admissibility relative to  $\mathcal{D}$  is not a decision of the Vortex. Any implementation may only expose structural relations for audit or visualization; no candidate trajectory receives authority from the Vortex.

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## 21. Relation to Epistemic Cryptography

Epistemic Cryptography may record structural quantities associated with  $\Phi$ ,  $R$ ,  $\Delta$ ,  $\mu$ ,  $A$ ,  $h$  or  $QE\mathcal{D}$ . However, it does not modify  $\Phi$ ,  $R$ ,  $\Delta$ ,  $\mathcal{D}$ , or enforce  $\kappa$  or decide  $QE$ . It only observes, records, hashes and preserves structural trace. Formally

$$\frac{\partial \Phi}{\partial \text{Audit}} = 0, \quad \frac{\partial \mathcal{D}}{\partial \text{Audit}} = 0$$

Audit has no command authority.

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## 22. Relation to Projection

Projection may expose structural features of  $\Phi$  or  $\Delta$  into a symbolic or linguistic layer. Projection does not interpret, decide or optimize. Projection does not write back into  $\Phi$  or alter  $\mathcal{D}$ . A projection may be empty; silence is valid. Formally

$$\Pi : \Phi \rightarrow \Gamma$$

where  $\Gamma$  is a projection space. The LLM adapter may render  $\Gamma$  into language

$$\text{LLM}_{\text{adapter}} : \Gamma \rightarrow \text{Language}$$

The LLM adapter is not a truth-bearing authority.

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## 23. Non-Algorithmic Closure

$\mathcal{D}$  must never be implemented as a classifier with authority, scoring function, reward model, safety filter, deployment gate, decision mechanism, optimization objective, runtime controller or agentic validator. Any implementation may only approximate structural membership for audit, visualization, documentation, deterministic consistency checking or non-authoritative analysis. Such approximation has no ontological authority; the ontology remains primary.

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## 24. Forbidden Interpretations

The following interpretations are incompatible with this anchor:

- $\mathcal{D}$  as algorithm
- $\mathcal{D}$  as safety filter
- $\mathcal{D}$  as classifier
- $\mathcal{D}$  as decision boundary
- $\text{Rep}(\Delta)$  as score or probability
- $K\mathcal{D}(\Phi)$  as metric, reward or tunable parameter
- $\kappa$  as threshold
- $QE\mathcal{D}$  as error or fallback
- Vortex as selector by  $\mathcal{D}$

Audit as enforcement  
 Projection as interpretation  
 LLM as authority  
 Memory as ontology

If any of these interpretations appear in downstream documentation or implementation, the downstream artefact is not compatible with this anchor.

## 25. Canonical Statement

$\mathcal{D}$  is the domain of epistemic curvature configurations that can still exist as meaningful relational structures.  $\kappa$  is the boundary of that representability.  $K\mathcal{D}(\Phi)$  is the curvature-domain predicate of representability.  $QE\mathcal{D}$  is the curvature-domain expression of non-representable aporia. Outside  $\mathcal{D}$  there is no prohibited state; there is no representable state.

## 26. Final Formal Closure

$$\Phi = (\Sigma, R)$$

$$R \in \mathfrak{so}(8), \quad R_{ij} = -R_{ji}, \quad R_{ii} = 0$$

$$\Delta = d_1 R$$

$$(d_1 R)_{ijk} = R_{ij} + R_{jk} + R_{ki}$$

$$d_2 d_1 = 0$$

$$\mathcal{D}_{\text{alg}} = \text{Im}(d_1)$$

$$\mathcal{D}_{\text{tri}} = \mathcal{D}_{\text{alg}} \cap \text{Fix}(\mathcal{T})$$

$$\mathcal{D} = \{\Delta \in C^2 \mid \Delta = d_1 R, d_2 \Delta = 0, P_{\mathcal{T}} \Delta = \Delta, \text{Rep}(\Delta) = 1\}$$

$$E = \{\Phi = (\Sigma, R) \mid d_1 R \in \mathcal{D}\}$$

$$\Xi = E = d_1^{-1}(\mathcal{D})$$

$$K_{\mathcal{D}}(\Phi) = 1 \iff d_1 R \in \mathcal{D}$$

$$K_{\mathcal{D}}(\Phi) = 0 \iff d_1 R \notin \mathcal{D}$$

$\mathcal{D}$

$$\kappa = \partial_{\mathcal{D}_{\text{tri}}} \mathcal{D}$$

$$QE_{\mathcal{D}} \iff d_1 R \notin \mathcal{D}$$

$\mathcal{D}$

## 27. End of Anchor

This document closes the formal curvature-domain expression of  $\mathcal{D}$ ,  $\Delta$ ,  $E$ ,  $\Xi$ ,  $K\mathcal{D}(\Phi)$ ,  $\kappa$  and  $QE\mathcal{D}$  for VECTAETOS 1.x. It does not redefine  $\Phi$ ,  $K(\Phi)$ ,  $\kappa$ ,  $QE$ , Vortex, audit, projection, memory or the LLM adapter outside their canonical roles. It specifies the admissible curvature-domain formalism.

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